RC CIRCULAR COLUMN CHECK – FLEXURE AND SHEAR (EC2)

1. DESIGN BASIS

EN 1992-1-1. Strips and bars, strain pivots A/B/C, P–M curve in the MEd direction.

Flexure: ray from 0 through (MEd, NEd) to the P–M curve (UR_M). fcd and shear (spirals):

EN 1992-1-1:2023 (5.3), §8.2.3.

N positive in compression. Annex A.n is calculated at its link ▸.

2. INPUTS

2.1 Materials

Characteristic concrete cylinder strength $f_{ck} := 35 \cdot \mathrm{MPa}$

Partial factor for concrete $\gamma_C := 1.5$

Sustained-load factor (5.3): 0.85, or 1.0 if loaded after 90 days $k_{tc} := 0.85$

Yield strength of longitudinal bars $f_{yk} := 500 \cdot \mathrm{MPa}$

Yield strength of spirals $f_{ywk} := 500 \cdot \mathrm{MPa}$

Ductility class of the spirals: 1 = class A, 0 = class B or C $\delta_A := 0$

Partial factor for reinforcement $\gamma_S := 1.15$

Elastic modulus of reinforcement $E_s := 200 \cdot \mathrm{GPa}$

Ultimate concrete strain (Table 3.1) $\varepsilon_{cu2} := 0.0035$

Concrete strain at peak stress $\varepsilon_{c2} := 0.002$

Design limit of steel strain, 0.9·εuk $\varepsilon_{ud} := 0.045$

2.2 Section and reinforcement sets

Column diameter $D := 1200 \cdot \mathrm{mm}$

Nominal cover to the set-1 spiral $c_{nom} := 75 \cdot \mathrm{mm}$

Reinforcement sets: 1 = outer, 2 = outer + inner $n_{set} := 1$

Set 1 (outer): longitudinal bar diameter $\phi_{l1} := 32 \cdot \mathrm{mm}$

Set 1: number of bars (evenly spaced) $n_{l1} := 22$

Set 1: spiral bar diameter $\phi_{w1} := 20 \cdot \mathrm{mm}$

Set 1: spiral pitch $s_1 := 85 \cdot \mathrm{mm}$

Set 2 (inner): longitudinal bar diameter $\phi_{l2} := 25 \cdot \mathrm{mm}$

Set 2: number of longitudinal bars $n_{l2} := 16$

Set 2: spiral bar diameter $\phi_{w2} := 16 \cdot \mathrm{mm}$

Set 2: spiral pitch $s_2 := 150 \cdot \mathrm{mm}$

2.3 Integration

Number of concrete strips across D $n_c := 100$

2.4 Design actions (ULS)

Axial force, compression positive $N_{Ed} := 8000 \cdot \text{kN}$

Bending moment about x $M_{Ed.x} := 3000 \cdot \text{kN} \cdot \text{m}$

Bending moment about y $M_{Ed.y} := 2000 \cdot \text{kN} \cdot \text{m}$

Shear force $V_{Ed} := 2500 \cdot \text{kN}$

3. DESIGN VALUES

Effective strength factor (5.4), fck,ref = 40 MPa $\eta_{cc} := \min\left(\left(\frac{40 \cdot \text{MPa}}{f_{ck}}\right)^{\frac{1}{3}}, 1\right) = 1$

Design compressive strength (5.3) $f_{cd} := \frac{\eta_{cc} k_{tc} f_{ck}}{\gamma_C} = 19.83 \cdot \text{MPa}$

Design yield strength of longitudinal bars $f_{yd} := \frac{f_{yk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strength of spirals $f_{ywd} := \frac{f_{ywk}}{\gamma_S} = 434.8 \cdot \text{MPa}$

Design yield strain of longitudinal bars $\varepsilon_{yd} := \frac{f_{yd}}{E_s} = 0.00217391$

Set-2 switch: 1 = included, 0 = ignored

$$\delta_2 := \begin{cases} 1 & \text{if } n_{set} \geq 2 \\ 0 & \text{else} \end{cases} = 0$$

4. SECTION GEOMETRY

4.1 Rings, bars and spirals

Section radius R (centroid to the outer face) $R := \frac{D}{2} = 600 \cdot \text{mm}$

Cover to set-2 spiral (inside set-1 bars) $c_2 := c_{nom} + \phi_{w1} + \phi_{l1} = 127 \cdot \text{mm}$

Radius to set-1 bar centres $r_1 := R - c_{nom} - \phi_{w1} - \frac{\phi_{l1}}{2} = 489 \cdot \text{mm}$

Radius to set-2 bar centres $r_2 := R - c_2 - \phi_{w2} - \frac{\phi_{l2}}{2} = 444 \cdot \text{mm}$

Set 2 must sit inside set 1 $r_2 = 444 \cdot \text{mm} \leq r_1 - \frac{\phi_{l1} + \phi_{l2}}{2} = 460 \cdot \text{mm}$ PASS

Area of one set-1 longitudinal bar $A_{b1} := 0.25\pi\phi_{l1}^2 = 804 \cdot \text{mm}^2$

Area of one set-2 longitudinal bar $A_{b2} := 0.25\pi\phi_{l2}^2 = 491 \cdot \text{mm}^2$

Gross section area $A_g := \pi R^2 = 1130973 \cdot \text{mm}^2$

Total longitudinal steel area $A_{st} := n_{l1} A_{b1} + \delta_2 n_{l2} A_{b2} = 17693 \cdot \text{mm}^2$

Spiral centreline diameter, set 1 $D_{sp.1} := D - 2c_{nom} - \phi_{w1} = 1030 \cdot \text{mm}$

Spiral centreline diameter, set 2 $D_{sp.2} := D - 2c_2 - \phi_{w2} = 930 \cdot \text{mm}$

Area of one set-1 spiral bar $A_{w1} := 0.25\pi\phi_{w1}^2 = 314 \cdot \text{mm}^2$

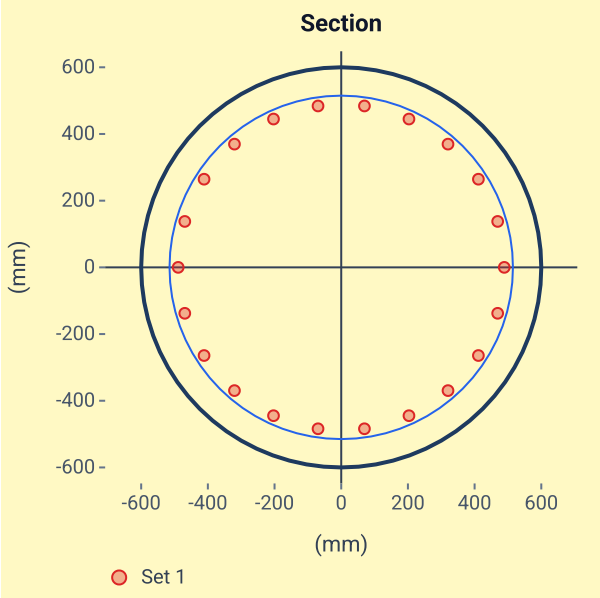
Area of one set-2 spiral bar $A_{w2} := 0.25\pi\phi_{w2}^2 = 201 \cdot \text{mm}^2$

Clear spacing of set-1 bars on the ring $s_{cl.1} := \frac{2\pi r_1}{n_{l1}} - \phi_{l1} = 108 \cdot \text{mm}$

Minimum clear spacing, EN 1536 bored piles $s_{\text{cl.1}} = 108 \cdot \text{mm} \geq 100 \cdot \text{mm}$ PASS

4.2 Bars and section

▸ **A.1** Bar coordinates · see page 6



4.3 Concrete strips

▸ **A.2** Concrete strips · see page 7

5. INTERACTION CURVE (N–M AT THE DEMAND DIRECTION)

5.1 Bending direction

Resultant design moment $M_{Ed} := \sqrt{M_{Ed.x}^2 + M_{Ed.y}^2} = 3605.6 \cdot \text{kN} \cdot \text{m}$

Direction of MEd, signs kept (0 ... 360°) $\theta_M := \text{angle}\left(M_{Ed.x}, M_{Ed.y}\right) = 33.7^\circ$

▸ **A.3** Strain plane, stress laws and section resultants · see page 7

5.2 Control points of the curve

The curve is evaluated at 55 neutral-axis depths from pure tension to uniform compression.

▸ **A.4** P–M diagram data (control points) · see page 8

Max. axial resistance (uniform εc2) $N_{Rd.max} = 29159.2 \cdot \text{kN}$

Maximum moment resistance on the curve $M_{Rd.max} = 4726.7 \cdot \text{kN} \cdot \text{m}$

6. FLEXURAL CHECK

Ray from the origin through the design point (MEd, NEd) to the P–M curve, as MBColumn.

6.1 Radial ratio

Normalised design axial force $$n_{Ed} := \frac{N_{Ed}}{N_{Rd.max}} = 0.27436$$

Normalised design moment $$m_{Ed} := \frac{M_{Ed}}{M_{Rd.max}} = 0.7628$$

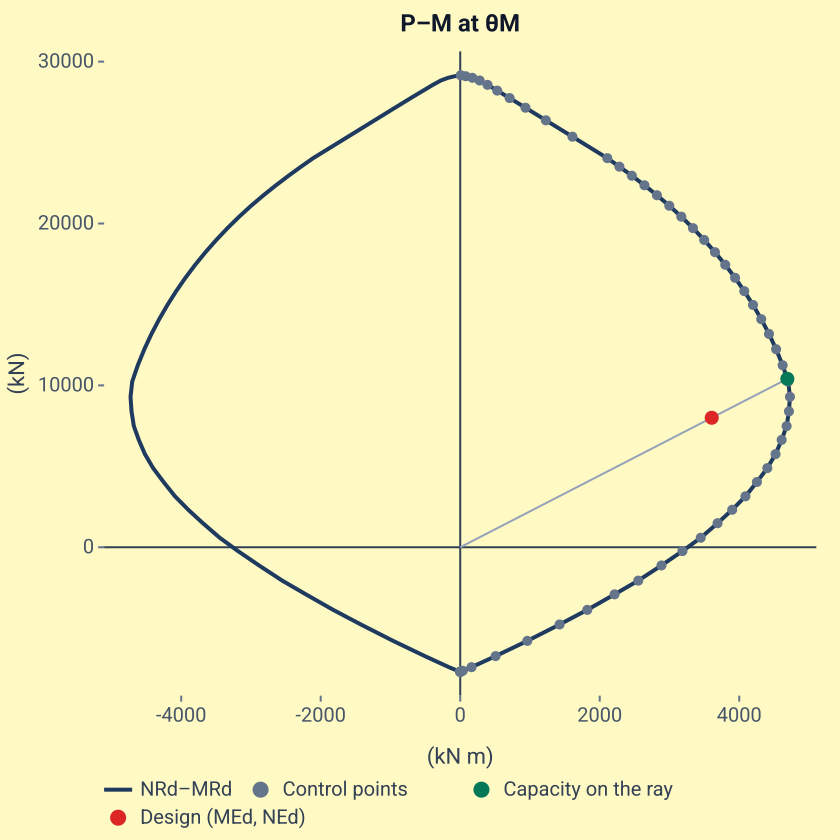
Distance of the design point from the origin $$\rho_{Ed} := \sqrt{n_{Ed}^2 + m_{Ed}^2} = 0.81064$$

Curve distance along the design ray $$\rho_{Rd} := \text{linterp}\left(s_v, \rho_v, \frac{n_{Ed}}{\rho_{Ed}}\right) = 1.0543$$

Radial ratio (scale-independent) $$\text{UR}_M := \frac{\rho_{Ed}}{\rho_{Rd}} = 0.7689$$

Check $$\text{UR}_M = 0.7689 \leq 1$$ PASS

6.2 P–M diagram at θM



7. SHEAR CHECK (EN 1992-1-1:2023 §8.2.3, SPIRALS)

7.1 Chords, web width and stresses

Truss in stresses: τEd = VEd/(bw·z) (8.18); spirals τRd,sy = ρw·fywd·cotθ (8.42); strut σcd = τEd·(cotθ + tanθ) ≤ ν·fcd (8.44), ν = 0.5 (8.2.3(6)). bw: least width between the compression chord (centroid of the concrete compression, §8.1) and the tension chord (d).

Each spiral crosses the shear plane twice: Asw = 2·Aw.

Effective depth: D/2 + half-ring centroid 2r/π, weighted by bar area

$$d := \frac{D}{2} + \frac{2}{\pi} \cdot \frac{n_{l1} A_{b1} r_1 + \delta_2 n_{l2} A_{b2} r_2}{n_{l1} A_{b1} + \delta_2 n_{l2} A_{b2}} = 911 \cdot \text{mm}$$

Lever arm, 8.2.1(3) $z := 0.9d = 820 \cdot \text{mm}$

Neutral-axis depth at the bending resistance under NEd (§8.1) $x := \text{linterp}\left(N_v, c_v, N_{Ed}\right) = 587 \cdot \text{mm}$

Depth of the compression chord centroid (concrete compression at x)

$$y_{cc} := \left\| \begin{array}{l} \text{if } x \le 0 \cdot \text{mm} \\ \quad \| 0 \cdot \text{mm} \\ \text{else} \\ \quad \left\| R - \frac{\text{sum}\left(\sigma_c\left(\varepsilon\left(x, x_c\right)\right) A_c x_c\right)}{\text{sum}\left(\sigma_c\left(\varepsilon\left(x, x_c\right)\right) A_c\right)} \right. \end{array} \right. = 286 \cdot \text{mm}$$

Web width: least chord width between the chords (only the tension chord if x ≤ 0)

$$b_w := \left\| \begin{array}{l} \text{if } x \le 0 \cdot \text{mm} \\ \quad \left\| 2\sqrt{R^2 - (d - R)^2} \right. \\ \text{else} \\ \quad \left\| \min\left(2\sqrt{R^2 - (R - y_{cc})^2}, 2\sqrt{R^2 - (d - R)^2}\right) \right. \end{array} \right. = 1022 \cdot \text{mm}$$

$$\tau_{Ed} := \frac{V_{Ed}}{b_w z} = 2.98 \cdot \text{MPa}$$

Average shear stress (8.18)

Strength reduction for cracked concrete, 8.2.3(6) $\nu := 0.5$

7.2 Strut angle

1 ≤ cotθ ≤ cot θmin (8.41): 2.5; 3.0 if σcp ≥ 3 MPa and x < 0.25d (linear between); 2.5 if x ≥ 0.25d; 2.5 − 0.1·|NEd|/VEd ≥ 1 in tension; × kA = 0.8 for class A spirals. cotθ < 1 (8.2.3(11)) is not used: σcd is least at θ = 45° and τRd,sy falls with cotθ.

Design cotθ: the largest value ≤ cot θmin that keeps σcd ≤ ν·fcd.

Factor on cot θmin: 0.8 for class A spirals, 8.2.3(4) $k_A := 1 - 0.2\delta_A = 1$

$$\sigma_{cp} := \frac{N_{Ed}}{A_g} = 7.07 \cdot \text{MPa}$$

Average axial stress, compression positive

Flattest strut allowed (8.41)

$$\cot_{\theta min} := \left\| \begin{array}{l} \text{if } N_{Ed} < 0 \cdot \text{kN} \\ \quad \left\| \max\left(1, k_A\left(2.5 - 0.1 \cdot \frac{|N_{Ed}|}{V_{Ed}}\right)\right) \right. \\ \text{else if } x \ge 0.25d \\ \quad \| 2.5 k_A \\ \text{else if } \sigma_{cp} \ge 3 \cdot \text{MPa} \\ \quad \| 3.0 k_A \\ \text{else} \\ \quad \left\| \left(2.5 + 0.5 \cdot \frac{\sigma_{cp}}{3 \cdot \text{MPa}}\right) k_A \right. \end{array} \right. = 2.5$$

Design cotθ: cot θmin if the strut holds, else the root of τEd(cotθ + tanθ) = ν·fcd; 1: strut fails

$$\cot_\theta := \begin{cases} \text{if } \tau_{Ed}\left(\cot_{\theta min} + \frac{1}{\cot_{\theta min}}\right) \le \nu f_{cd} \\ \quad \cot_{\theta min} \\ \text{else if } 2\tau_{Ed} \le \nu f_{cd} \\ \quad \left\| \frac{\nu f_{cd}}{2\tau_{Ed}} + \sqrt{\left(\frac{\nu f_{cd}}{2\tau_{Ed}}\right)^2 - 1} \right. \\ \text{else} \\ \quad 1 \end{cases} = 2.5$$

7.3 Resistance

Shear reinforcement ratio (8.43), set 2 when δ2 = 1 $\rho_w := \frac{2}{b_w}\left(\frac{A_{w1}}{s_1} + \delta_2 \frac{A_{w2}}{s_2}\right) = 0.00722992$

Spiral resistance (8.42) $\tau_{Rd.sy} := \rho_w f_{ywd} \cot_\theta = 7.86 \cdot \text{MPa}$

Strut stress (8.44) $\sigma_{cd} := \tau_{Ed}\left(\cot_\theta + \frac{1}{\cot_\theta}\right) = 8.65 \cdot \text{MPa}$

Shear utilisation ratio $\text{UR}_V := \max\left(\frac{\tau_{Ed}}{\tau_{Rd.sy}}, \frac{\sigma_{cd}}{\nu f_{cd}}\right) = 0.8718$

Check $\text{UR}_V = 0.8718 \le 1$ PASS

8. SUMMARY

Flexure utilisation ratio $\text{UR}_M = 0.7689$

Shear utilisation ratio $\text{UR}_V = 0.8718$

ANNEX

A.1 Bar coordinates

Bar index, set 1 (edit seq to renumber) $i_1 := \text{seq}\,(0, n_{l1} - 1)$

Bar index, set 2 $i_2 := \text{seq}\,(0, n_{l2} - 1)$

Angles for drawing circles $\varphi_t := \text{seq}\left(0, 2\pi, \frac{\pi}{36}\right)$

Set-1 bar coordinates, x $x_{l1} := r_1 \cos\left(\frac{2\pi i_1}{n_{l1}}\right)$

Set-1 bar coordinates, y $y_{l1} := r_1 \sin\left(\frac{2\pi i_1}{n_{l1}}\right)$

Set-2 bar coordinates, x $x_{l2} := r_2 \cos\left(\frac{2\pi i_2}{n_{l2}}\right)$

Set-2 bar coordinates, y $y_{l2} := r_2 \sin\left(\frac{2\pi i_2}{n_{l2}}\right)$

Set 1 (outer)

$$i_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \\ 15 \\ 16 \\ 17 \\ 18 \\ 19 \\ 20 \\ 21 \end{bmatrix} \quad x_{l1} = \begin{bmatrix} 489 \\ 469 \\ 411 \\ 320 \\ 203 \\ 70 \\ -70 \\ -203 \\ -320 \\ -411 \\ -469 \\ -489 \\ -469 \\ -411 \\ -320 \\ -203 \\ -70 \\ 70 \\ 203 \\ 320 \\ 411 \\ 469 \end{bmatrix} \text{ mm} \quad y_{l1} = \begin{bmatrix} 0 \\ 138 \\ 264 \\ 370 \\ 445 \\ 484 \\ 484 \\ 445 \\ 370 \\ 264 \\ 138 \\ 0 \\ -138 \\ -264 \\ -370 \\ -445 \\ -484 \\ -484 \\ -445 \\ -370 \\ -264 \\ -138 \end{bmatrix} \text{ mm}$$

Set 2 (inner, used when nset = 2)

$$i_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ 12 \\ 13 \\ 14 \\ 15 \end{bmatrix} \quad x_{l2} = \begin{bmatrix} 444 \\ 411 \\ 314 \\ 170 \\ 0 \\ -170 \\ -314 \\ -411 \\ -444 \\ -411 \\ -314 \\ -170 \\ 0 \\ 170 \\ 314 \\ 411 \end{bmatrix} \text{ mm} \quad y_{l2} = \begin{bmatrix} 0 \\ 170 \\ 314 \\ 411 \\ 444 \\ 411 \\ 314 \\ 170 \\ 0 \\ -170 \\ -314 \\ -411 \\ -444 \\ -411 \\ -314 \\ -170 \end{bmatrix} \text{ mm}$$

A.2 Concrete strips

The concrete is cut into nc thin strips parallel to the neutral axis. Each strip carries the stress at its centre strain; summing strip forces (× lever arm) gives the concrete N and M.

Strip index $j := \text{seq}(0, n_c - 1)$

Strip width $\Delta := \dfrac{D}{n_c} = 12 \cdot \text{mm}$

Strip centre coordinate, from the centroid $x_c := R - (j + 0.5)\,\Delta$

Strip area (chord length × width) $A_c := 2\sqrt{R^2 - x_c^2} \cdot \Delta$

Strip area / gross area (check of nc) $\dfrac{\text{sum}(A_c)}{A_g} = 1.0003$

A.3 Strain plane, stress laws and section resultants

Set-1 bar coordinate p along θM (from centroid) $p_{l1} := r_1 \cos\left(\dfrac{2\pi i_1}{n_{l1}}\,\text{rad} - \theta_M\right)$

Set-2 bar coordinate p along θM $p_{l2} := r_2 \cos\left(\dfrac{2\pi i_2}{n_{l2}}\,\text{rad} - \theta_M\right)$

Depth dθ: compression edge to extreme bar

$$d_\theta := \begin{cases} \text{if } n_{set} \geq 2 \\ \quad \| R - \min(p_{l1}, p_{l2}) \\ \text{else} \\ \quad \| R - \min(p_{l1}) \end{cases} = 1089 \cdot \text{mm}$$

Depth cA at pivot A/B boundary $c_A := \dfrac{\varepsilon_{cu2} d_\theta}{\varepsilon_{cu2} + \varepsilon_{ud}} = 79 \cdot \text{mm}$

Depth yC of pivot C (εc2 in full compression) $y_C := \left(1 - \dfrac{\varepsilon_{c2}}{\varepsilon_{cu2}}\right) D = 514 \cdot \text{mm}$

Strain plane (+ compression)

c = neutral-axis depth below the compression edge; p = fibre coordinate from the centroid along θM, so R − p = fibre depth. The plane pivots about A (extreme bar at −εud, c < cA), B (edge at εcu2, c ≤ D) or C (εc2 at depth yC, c > D), EC2 Fig. 6.1.

Strain at fibre p for depth c

$$\varepsilon(c,p) := \left\| \begin{array}{l} \text{if } c < c_A \\ \quad \left\| \varepsilon_{ud}\,\dfrac{c-R+p}{d_\theta - c} \right. \\ \text{else if } c \le D \\ \quad \left\| \varepsilon_{cu2}\,\dfrac{c-R+p}{c} \right. \\ \text{else} \\ \quad \left\| \varepsilon_{c2}\,\dfrac{c-R+p}{c-y_C} \right. \end{array} \right.$$

Stress laws

Concrete stress, parabola-rectangle (EC2 3.17), no tension

$$\sigma_c(\varepsilon) := \left\| \begin{array}{l} \text{if } \varepsilon \le 0 \\ \quad \| 0 \cdot \mathrm{MPa} \\ \text{else if } \varepsilon < \varepsilon_{c2} \\ \quad \left\| f_{cd}\left(1 - \left(1 - \dfrac{\varepsilon}{\varepsilon_{c2}}\right)^2\right) \right. \\ \text{else} \\ \quad \| f_{cd} \end{array} \right.$$

Steel stress, elastic-plastic within ±fyd

$$\sigma_s(\varepsilon) := \left\| \begin{array}{l} \text{if } E_s\varepsilon < -f_{yd} \\ \quad \| -f_{yd} \\ \text{else if } E_s\varepsilon > f_{yd} \\ \quad \| f_{yd} \\ \text{else} \\ \quad \| E_s\varepsilon \end{array} \right.$$

Net bar stress (concrete it displaces subtracted)

$$\sigma_b(\varepsilon) := \sigma_s(\varepsilon) - \sigma_c(\varepsilon)$$

Section resultants

Concrete strips (xc, Ac) plus bars of set 1 and, when δ2 = 1, set 2:

Axial resistance for depth c

$$N_R(c) := \mathrm{sum}\left(\sigma_c\left(\varepsilon\left(c, x_c\right)\right) A_c\right) + A_{b1}\mathrm{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l1}\right)\right)\right) + \delta_2 A_{b2}\mathrm{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l2}\right)\right)\right)$$

Moment resistance for depth c (about the centroid)

$$M_R(c) := \mathrm{sum}\left(\sigma_c\left(\varepsilon\left(c, x_c\right)\right) A_c x_c\right) + A_{b1}\mathrm{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l1}\right)\right) p_{l1}\right) + \delta_2 A_{b2}\mathrm{sum}\left(\sigma_b\left(\varepsilon\left(c, p_{l2}\right)\right) p_{l2}\right)$$

A.4 P–M diagram data (control points)

Neutral-axis depths: −0.1D (all bars yielded in tension) … 100D (uniform εc2)

$$c_v := \mathrm{stack}\left(\mathrm{seq}\left(-0.1, 1, 0.025\right) D,\ \frac{D}{\mathrm{seq}\left(0.9, 0.1, -0.1\right)},\ 100D\right)$$

Number of control points

$$\mathrm{len}\left(c_v\right) = 55$$

Strain at the compression edge for each point

$$\varepsilon_{top} := \mathrm{map}\left(\varepsilon, c_v, R\right)$$

Strain at the extreme tension bar for each point $\varepsilon_{bar} := \mathrm{map}\left(\varepsilon, c_v, R - d_\theta\right)$

Axial resistance at each point $N_v := \mathrm{map}\left(N_R, c_v\right)$

Moment resistance at each point $M_v := \mathrm{map}\left(M_R, c_v\right)$

Control points at θM (N compression +)

$c_v =$ mm

rows 1:28	rows 29:55
−120	720
−90	750
−60	780
−30	810
0	840
30	870
60	900
90	930
120	960
150	990
180	1020
210	1050
240	1080
270	1110
300	1140
330	1170
360	1200
390	1333
420	1500
450	1714
480	2000
510	2400
540	3000
570	4000
600	6000
630	12000
660	120000
690	

$\varepsilon_{top} =$

rows 1:28	rows 29:55
−0.0044668	0.0035
−0.0034353	0.0035
−0.00235	0.0035
−0.0012065	0.0035
0	0.0035
0.0012749	0.0035
0.0026241	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0035
0.0035	0.0032558
0.0035	0.0030435
0.0035	0.0028571
0.0035	0.0026923
0.0035	0.0025455
0.0035	0.0024138
0.0035	0.0022951
0.0035	0.0021875
0.0035	0.0020896
0.0035	0.0020086
0.0035	

$\varepsilon_{bar} =$

rows 1:28	rows 29:55
−0.045	−0.0017934
−0.045	−0.0015817
−0.045	−0.0013862
−0.045	−0.0012053
−0.045	−0.0010372
−0.045	−0.00088076
−0.045	−0.00073473
−0.038847	−0.00059813
−0.02826	−0.00047006
−0.021908	−0.00034976
−0.017674	−0.00023653
−0.014649	−0.00012977
−0.01238	-2.8943×10^{-5}
−0.010616	6.6434×10^{-5}
−0.0092042	0.00015679
−0.0080493	0.00024251
−0.0070868	0.00032395
−0.0062725	0.0005968
−0.0055744	0.00083405
−0.0049695	0.0010423
−0.0044401	0.0012264
−0.0039731	0.0013905
−0.0035579	0.0015376
−0.0031864	0.0016703
−0.0028521	0.0017905
−0.0025496	0.0018999
−0.0022746	0.0019904
−0.0020236	

$N_v =$ rows 1:28 rows 29:55 kN $M_v =$ rows 1:28 rows 29:55 kN · m

N_v rows 1:28	N_v rows 29:55	M_v rows 1:28	M_v rows 29:55
−7693	12235	0	4528
−7693	13180	0	4426
−7693	14091	0	4315
−7693	14969	0	4197
−7693	15821	0	4072
−7631	16647	36	3940
−7410	17451	162	3799
−6725	18229	507	3652
−5783	18984	962	3497
−4771	19714	1424	3337
−3868	20423	1821	3169
−2911	21099	2212	2997
−2061	21747	2552	2820
−1122	22363	2885	2641
−236	22952	3186	2461
587	23511	3448	2282
1486	24037	3689	2108
2317	25364	3899	1608
3147	26375	4089	1228
4032	27153	4253	935
4889	27751	4402	708
5754	28210	4520	530
6632	28568	4608	391
7483	28834	4679	278
8396	29002	4712	174
9294	29100	4727	78
10239	29159	4702	7
11245		4622	

Max. axial resistance (uniform εc2) $N_{Rd.max} := \max(N_v) = 29159.2 \cdot \text{kN}$

Maximum moment resistance on the curve $M_{Rd.max} := \max(M_v) = 4726.7 \cdot \text{kN} \cdot \text{m}$

Normalised N of the control points

$$n_v := \frac{N_v}{N_{Rd.max}}$$

Normalised M of the control points

$$m_v := \frac{M_v}{M_{Rd.max}}$$

Distance of each control point from the origin $\rho_v := \sqrt{n_v^2 + m_v^2}$

Direction of each point (sine of angle)

$$s_v := \frac{n_v}{\rho_v}$$